

SECTION 05 12 00

STRUCTURAL STEEL FRAMING

Riverside Mixed-Use Development
Project No. RMD-2025-001

PART 1 - GENERAL

1.01 REFERENCES

- AISC 360 - Specification for Structural Steel Buildings
- AISC 341 - Seismic Provisions for Structural Steel Buildings
- AWS D1.1 - Structural Welding Code - Steel
- ASTM A992 - Structural Steel Shapes
- RCSC - Specification for Structural Joints Using High-Strength Bolts
- ASTM A500 - Cold-Formed Welded Structural Hollow Sections
- SSPC - Surface Preparation Standards

1.02 SUBMITTALS

- A. Shop drawings showing member sizes, connection details, bolt patterns, and weld symbols.
- B. Mill certificates for all structural steel shapes and plates.
- C. Welding procedure specifications (WPS) per AWS D1.1.
- D. Bolt installation procedures including pre-installation verification testing.
- E. Erection plan including crane locations, pick sequences, and temporary bracing.

1.03 QUALITY ASSURANCE

- A. Fabricator: AISC certified, Category STD (Standard for Steel Building Structures).
- B. Erector: AISC certified, Category CSE (Certified Steel Erector).
- C. Welders: AWS certified per D1.1 for required positions and processes.

PART 2 - PRODUCTS

2.01 STRUCTURAL STEEL

Shape Type	ASTM Standard	Grade	Fy (ksi)	Fu (ksi)
Wide Flange	A992	50	50	65
HSS Round/Rect	A500	Grade C	50	62
Plates	A572	Grade 50	50	65
Angles/Channels	A36	-	36	58
Anchor Rods	F1554	Grade 55	55	75

All steel to be domestic (melted and manufactured in USA).

2.02 BOLTS AND CONNECTORS

- A. High-strength bolts: ASTM F3125, Grade A325 or A490.
- B. Shear studs: ASTM A108, AWS D1.1 Type B, 3/4-inch diameter x 4-1/2 inch.
- C. Direct tension indicators: ASTM F959.
- D. All connections slip-critical unless noted otherwise on drawings.

2.03 WELDING MATERIALS

- A. Electrodes: AWS A5.1 E70XX (SMAW); AWS A5.20 E71T-1 (FCAW).
- B. Weld metal: matching or overmatching base metal strength.
- C. Preheat per AWS D1.1 Table 3.2 based on thickness and grade.

2.04 FIREPROOFING

- A. SFRM: spray-applied fire-resistive material, UL listed.
- B. Columns: 2-hour fire rating. Beams and girders: 1.5-hour fire rating.
- C. Manufacturer: Carbolite Pyrocrete 241 or approved equal.
- D. Density: minimum 15 pcf for columns, 12 pcf for beams.

2.05 SHOP PRIMER

- A. Surface preparation: SSPC-SP6 Commercial Blast Cleaning.
- B. Primer: organic zinc-rich per SSPC Paint 20, 3.0 mils DFT.
- C. Do not prime surfaces to receive SFRM or concrete.

PART 3 - EXECUTION

3.01 FABRICATION

- A. Fabricate per AISC Code of Standard Practice.
- B. Tolerances: AISC Code of Standard Practice, Chapter 6.
- C. Mill scale removal at faying surfaces of slip-critical connections.

3.02 ERECTION

- A. Erect per AISC Code of Standard Practice.
- B. Plumbing tolerance: 1/500 of column height, maximum 1 inch.
- C. Temporary bracing per erection engineer's design.
- D. Do not field weld without approval of structural engineer of record.

3.03 BOLTING

- A. Slip-critical connections: turn-of-nut method per RCSC.
- B. Pre-installation verification testing required for each bolt lot.
- C. Inspection: verify 10% of bolts per connection using calibrated wrench.

3.04 QUALITY CONTROL

- A. Special inspection per IBC Section 1705.
- B. Ultrasonic testing of 100% CJP welds per AWS D1.1.
- C. Visual inspection of all welds per AWS D1.1.
- D. High-strength bolt installation verification per RCSC.
- E. Survey control: verify member elevations at each level before proceeding.

END OF SECTION 05 12 00